

Adaptive Alpha

INSTITUTIONAL SCREENER PLATFORM

User Guide & Methodology Manual

DOCUMENT VERSION

v8.1 (Enterprise)

CALCULATION METHOD

ARS / SRS Matrix

TARGET MARKET

NSE India (Nifty 200)

1. Platform Overview

Adaptive Alpha is a premium technical screener dashboard engineered to identify market leaders, breakouts, and sector rotation trends within the NSE India universe. The platform utilizes advanced price-volume calculation models to analyze momentum relative to the broader market index (NIFTY 50).

By shifting from standard trend metrics to adaptive relative strength, the application filters out market noise and equips analysts with high-probability long candidates matching institutional accumulation models.

Institutional Philosophy

Institutional capital does not buy stocks in isolation; it accumulates assets showing relative strength vs a benchmark index. Adaptive Alpha calculates this relationship mathematically, highlighting where "smart money" is actively rotating.

2. Core Technical Metrics

The dashboard renders six core technical metrics for each stock, aligning exactly with the underlying momentum mathematical algorithm:

Adaptive Relative Strength (ARS)

Unlike standard relative strength, ARS measures the asset's performance relative to the NIFTY 50 Index over a custom lookback window starting from a historical base date (Jan 1, 2021). It represents the percentage outperformance or underperformance relative to the benchmark:

$$\text{ARS} = ((\text{Stock Today} / \text{Stock Base Date}) / (\text{NIFTY Today} / \text{NIFTY Base Date})) - 1$$

- **ARS > 15% (Vibrant Green):** Strong institutional outperformance and leading market trend.
- **ARS > 5% (Soft Green):** Moderate outperformance / active accumulation.
- **ARS > 0% (Sage Green):** Mild outperformance / positive bias.
- **ARS > -5% (Muted Coral):** Mild underperformance / consolidating range.
- **ARS < -5% (Vibrant Red):** Strong underperformance / lagging trend.

Shorter-term Relative Strength (SRS)

SRS isolates near-term strength by calculating the relative performance over a fixed rolling window of **63 trading days** (approximately 3 calendar months). This helps spot near-term leaders before they reflect in the long-term ARS metric.

Volume Ratio

Measures the current day's trading volume against its **20-day simple moving average**. A value of 2.0x indicates the stock traded double its typical daily volume, signaling active institutional positioning.

52-Week Proximity (52W Prox)

Shows how close the stock's closing price is to its 52-week high. Values near 0.00% mean the stock is trading right at its 52-week high, which is typical for explosive momentum breakouts.

Days in Trend

The consecutive count of days the ARS metric has maintained its current sign (positive or negative). A sudden change indicates a momentum flip date.

Supertrend Indicator (Dual Parameters)

Supertrend is a trend-following indicator based on Average True Range (ATR) and median prices, serving as the second stage of our 2-stage screener. The platform supports two configurations:

- **Supertrend 14/3 (Standard):** Utilizes a 14-period ATR and 3x multiplier. This is the daily standard configuration, offering a smoother trend line and fewer whipsaws.
- **Supertrend 10/3 (Fast):** Utilizes a 10-period ATR and 3x multiplier. This is a more sensitive, faster-reacting trend line to capture early trend reversals.

A stock generates a **BUY** signal when the price closes above the trailing Supertrend line. A fresh trend flip is highlighted with a pulse glow and a ⚡ icon.

Relative Strength (RS) Vector Visualizer

The sidebar features a custom visual vector charting the stock's momentum in a 2-dimensional space, comparing its long-term strength (ARS) against its short-term rotation (SRS):

- **Horizontal Axis (X-Axis):** Long-term outperformance (ARS). Positive space (right) is leading, negative space (left) is lagging.
- **Vertical Axis (Y-Axis):** Shorter-term quarterly rotation (SRS). Positive space (up) shows increasing near-term strength, negative space (down) shows near-term consolidation.
- **The Vector Line:** Represents the momentum path starting from the Nifty 50 benchmark (the center grid crosshair) to the stock's current coordinates.
- **Interpretation:**
 - **Top-Right (Leading):** Strongly outperforming in both timeframes. Prime buy targets.
 - **Bottom-Right (Weakening):** Strong long-term but undergoing a short-term correction (represents low-risk pullback entries).
 - **Top-Left (Improving):** Weak long-term but starting to pick up near-term momentum (potential early turnarounds).
 - **Bottom-Left (Lagging):** Underperforming in both horizons. Generally avoided.

3. Sector Rotation (RRG Model)

The **Sector View** tab categorizes industries into four distinct momentum quadrants, inspired by Relative Rotation Graphs (RRG). This classification helps determine whether an industry is in favor or starting to lag:

Rotation Quadrant	Momentum Description	Institutional Bias	Visual Color
LEADING	High relative strength + positive momentum	Aggressive buying / Overweight	Emerald Green
WEAKENING	High relative strength but momentum is slowing	Hold / Trimming positions	Soft Amber
IMPROVING	Negative relative strength but momentum is turning positive	Accumulating / Early Watchlist	Soft Teal
LAGGING	Negative relative strength + negative momentum	Distribution / Underweight	Soft Coral

4. Setup Scanner Strategies

You can use the dashboard filters to isolate specific institutional trading setups:

Setup A: The Momentum Leader (Trend Following)

Targeting strong stocks that are steadily outperforming the market during a bull cycle.

- **Filter Configuration:** ARS > 0% (ON), ARS Trending ↑ (ON), Pass Only (ON), Group by Industry (ON).
- **Sorting:** Sort by *ARS High* → *Low*. Focus on the top 2 sectors with the highest passing rates.

Setup B: High-Volume Institutional Breakout

Identifying base breakouts accompanied by heavy institutional volume surges.

- **Filter Configuration:** ARS Trending ↑ (ON), VOL+ (ON), Near 52W High (ON).
- **Scanning:** Watch for stocks marked with the yellow ***NEW*** badge. This badge signals that the stock's ARS turned positive today for the first time in the lookback window.

Setup C: The 2-Stage Momentum & Supertrend Screener

Combining structural relative strength (long-term trend) with a short-term trend-following trigger to enter high-probability positions.

- **Filter Configuration:** ARS > 0% (ON), Supertrend BUY (ON), Pass Only (ON).
- **Param Selector:** Toggle between Supertrend 14/3 (Daily standard) and Supertrend 10/3 (Fast, sensitive) based on market volatility.
- **Scanning:** Scan for stocks that have strong baseline momentum (ARS > 0%) and have just flipped to a Supertrend BUY signal (indicated by the ⚡ icon on the BUY badge). This represents a low-risk pullback entry in a strong stock.

Volume Threshold Warning

Ensure the Volume Ratio is > 1.50x. Institutional buying leaves footprints—if a stock is breaking out to a 52-week high but volume is below average (ratio < 1.00x), the breakout is highly likely to fail.

5. Engine Data & TradingView Integration

Hybrid Data Architecture

The dashboard is powered by a dual-engine data architecture designed for instant access and accuracy:

1. **Precalculated closing database:** The system automatically fetches a daily pre-calculated dataset generated at the close of every market day. This downloads in under 50ms on startup.
2. **Intraday Live Scan:** Clicking the "🔄 Live Data" button fetches current intraday prices directly from Yahoo Finance and recalculates all momentum metrics in the browser.

One-Click Charts

To inspect a setup visually, click anywhere on a stock row. The platform will open the daily TradingView chart for that stock in a new tab: <https://www.tradingview.com/chart/?symbol=NSE:SYMBOL>

6. Support and Operations

The data pipeline updates automatically after market hours on trading days. If the daily GitHub Actions pipeline runs into formatting changes on the NSE exchange servers, it uses fallback parsing to ensure data continuity.

For custom additions to the constituent list (adding new symbols or modifying sector weights), developers can update the universe arrays directly inside the core scripts.

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